The Tarik Skubal Decision

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The Skubal Dilemma

- The Cubs season has just recently ended by getting swept in the Wild Card Series by the San Diego Padres
- The starting pitching was questionable for the majority of the season, and it continued in the playoffs
- Skubal is one of the top arms in the MLB who will be a free agent this offseason

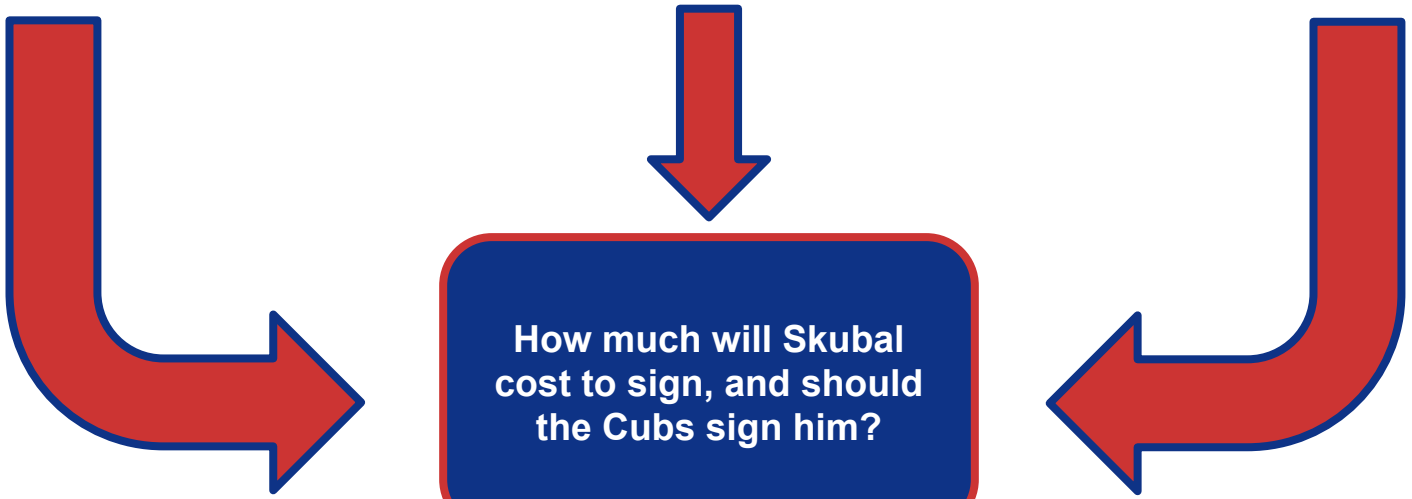


Allocation of Resources

Cubs are currently projected to be 12th in payroll next year

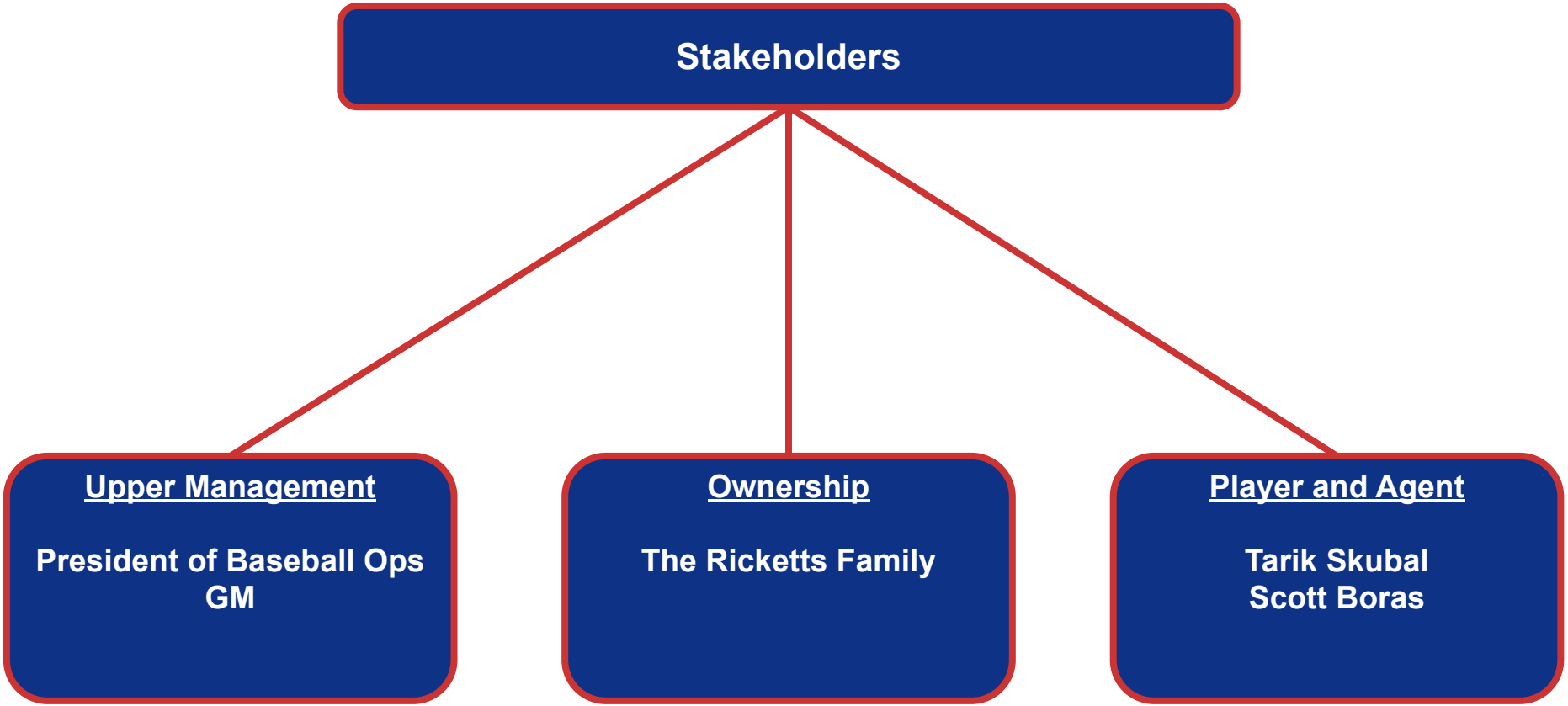
Starters of Skubal's caliber generally command north of $30 million per year

While a large market team that will spend, they can not afford to have Skubal underperform



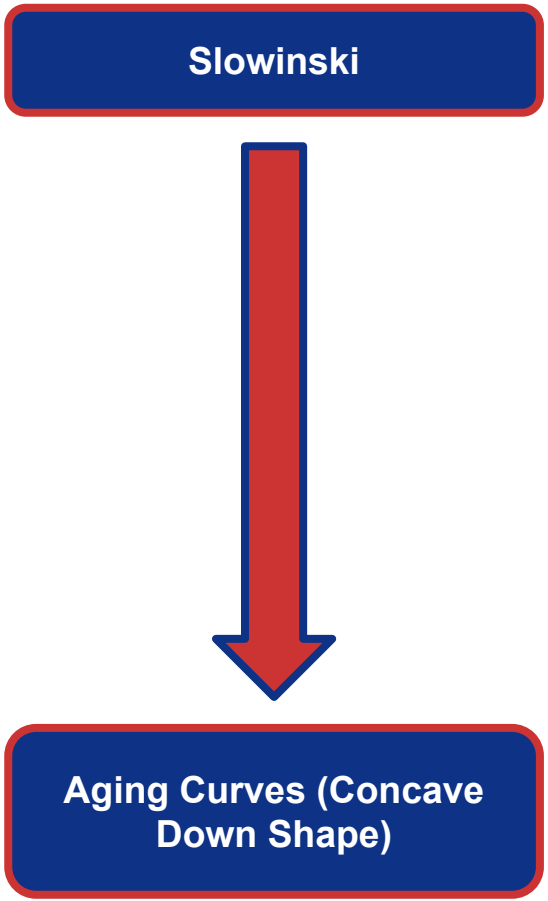
How much will Skubal cost to sign, and should the Cubs sign him?

Stakeholders



Prior Work


Clemens
WAR (Wins Above Replacement)
Dollars ($8mil / WAR)


Slowinski
Aging Curves (Concave Down Shape)

Dataset - Source and Structure

- Data originates from pybaseball which pulls from trusted MLB data sites
- Pitcher-season level data (2016-2026)
- Filtered for pitcher's who started 5 games
- 15 columns, 1791 rows







Dataset - Variables

Identifier

Player Name
Position
Throws
Season

Context

Age

Result Metrics

ERA
WHIP
FIP

Variables

Rate Metrics

K%
BB%

Valuation Metrics (response)

WAR
Dollars

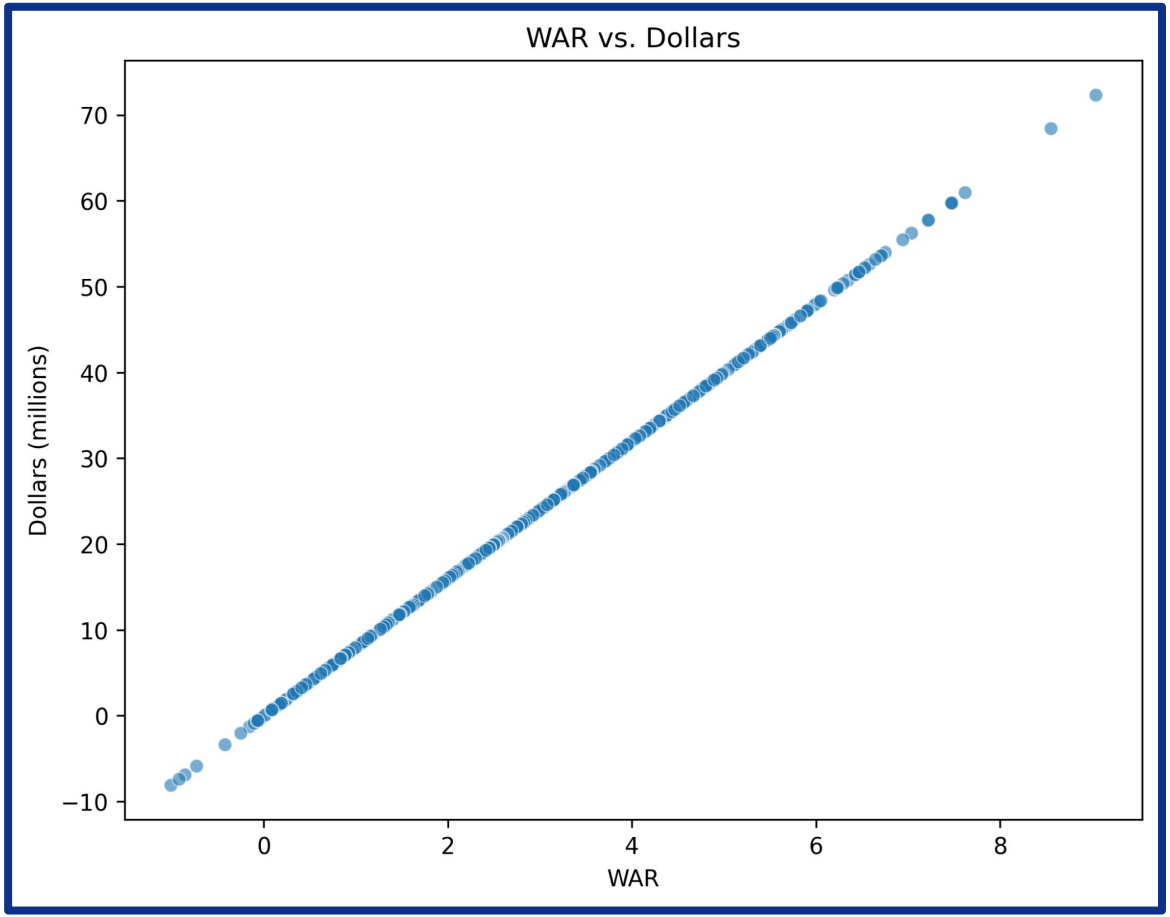
Velocity and Contact

FB Velocity
Stuff+
Barrel%

Exploratory Data Analysis

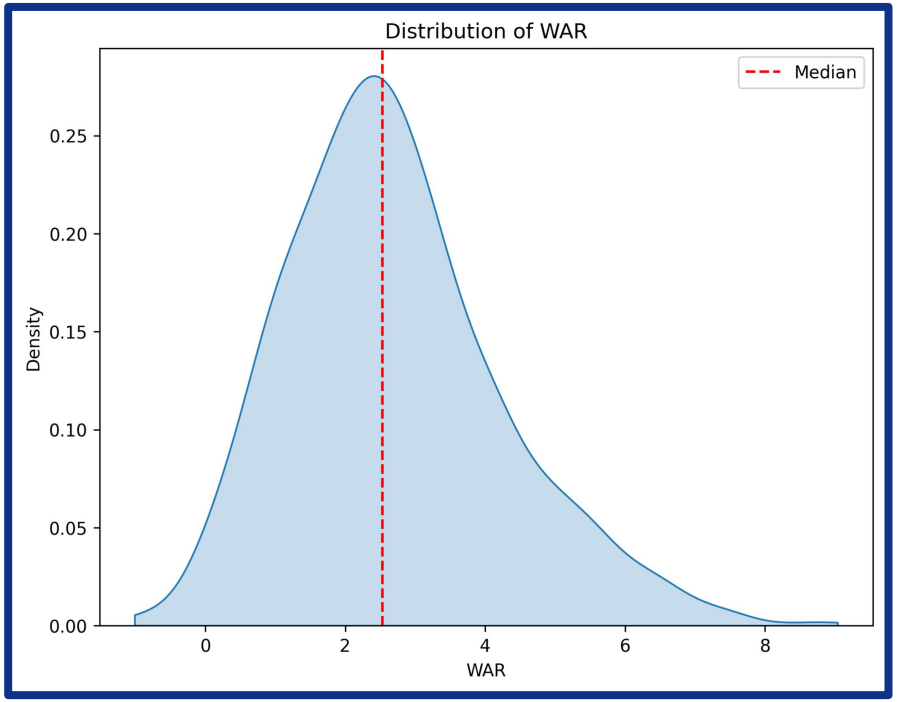
Relationship between WAR and Dollars

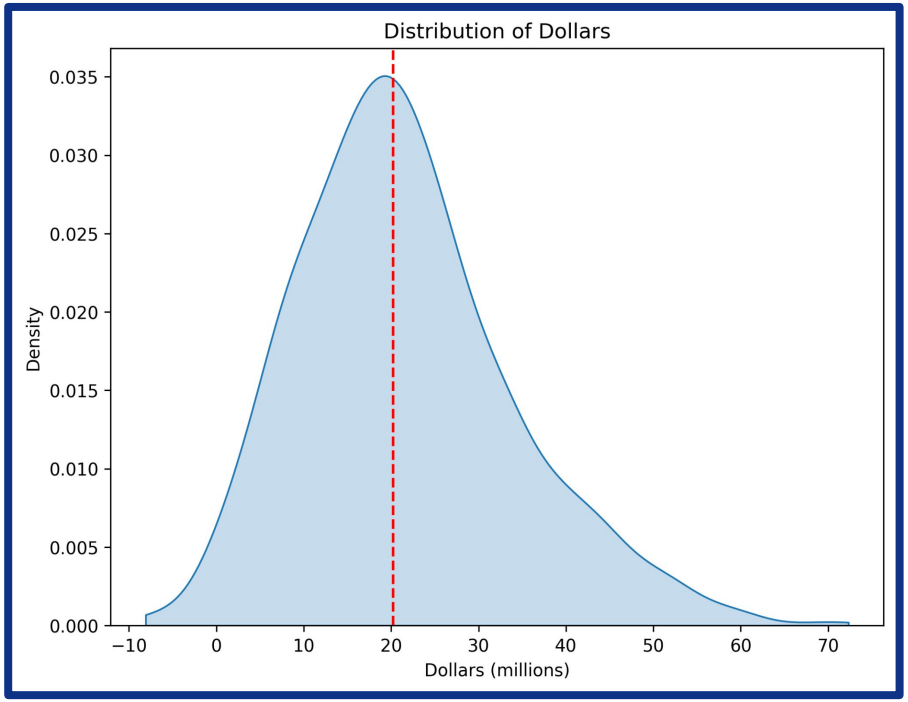
- WAR is perfectly linear with Dollars as Dollars is calculated directly from WAR
- One will be the target variable while the other will have to be completely left out of the model



Distribution of WAR and Dollars

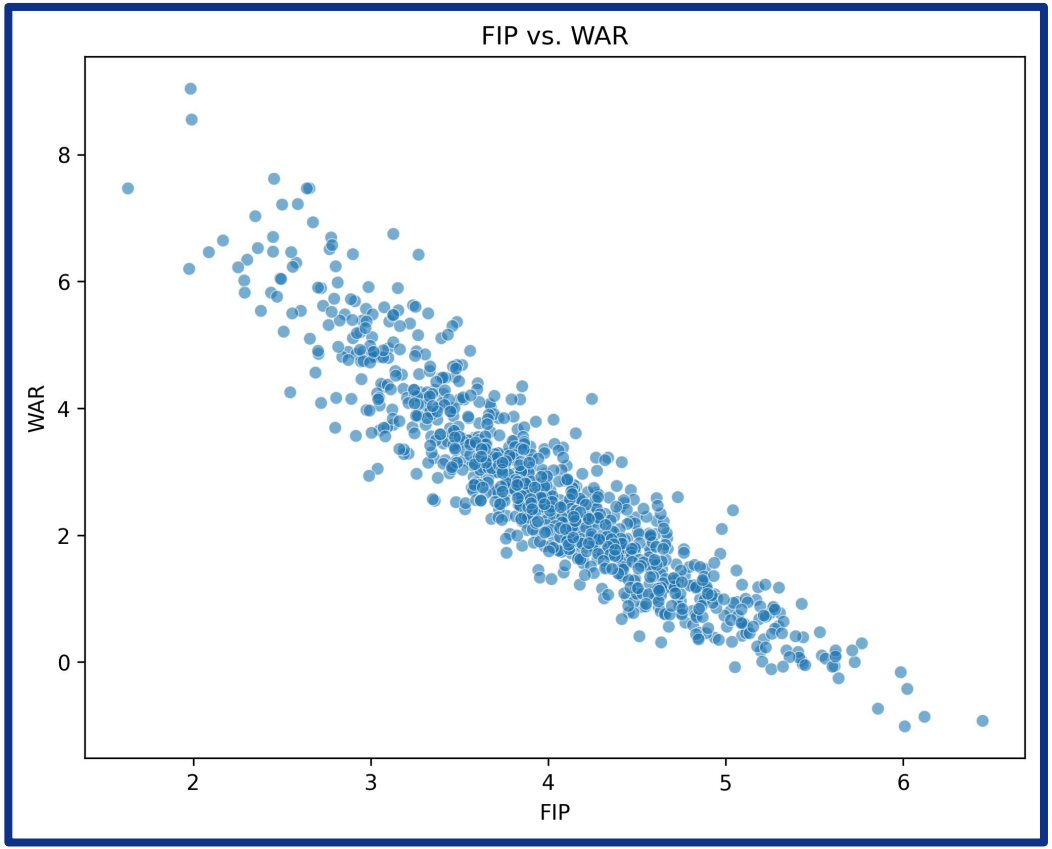
Given that both have a perfect linear relationship, they should have the exact same distribution shape. Both distributions are skewed to the right.





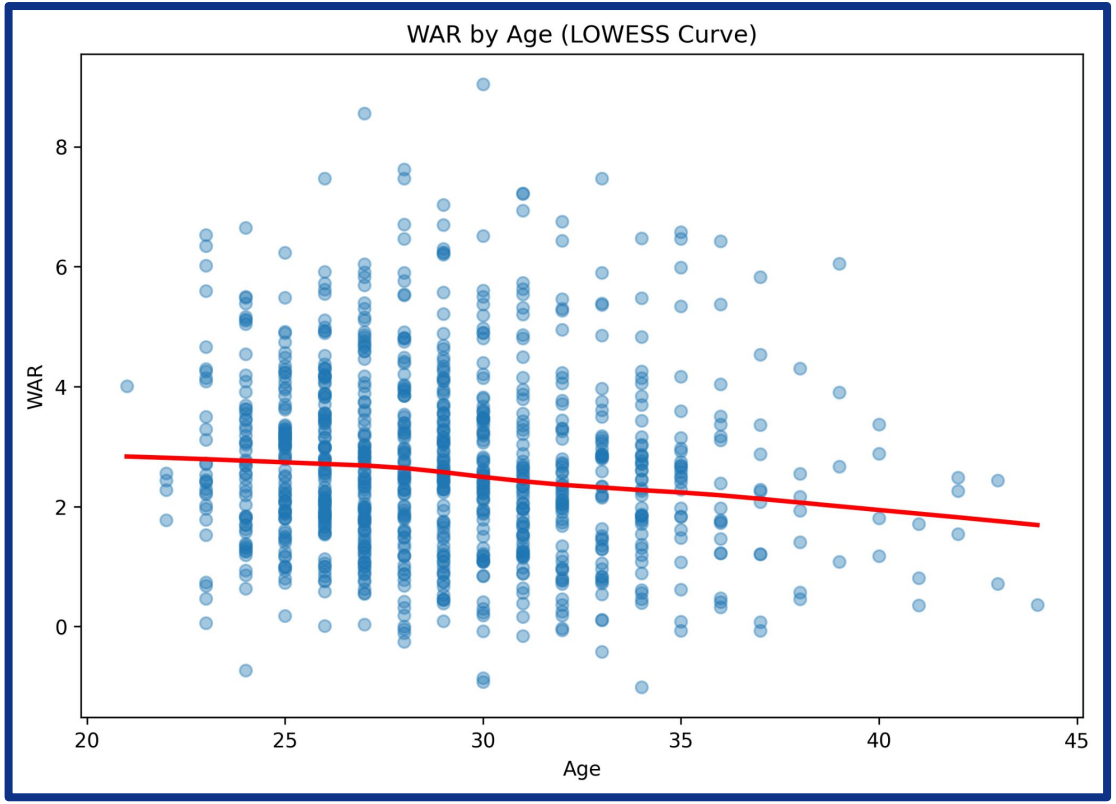
FIP vs WAR

- FIP looks at the quality of a pitcher without factoring the team's defense
  - Lower FIP = Better performance
- Skubal's FIP this year was 2.51 making him harder to predict



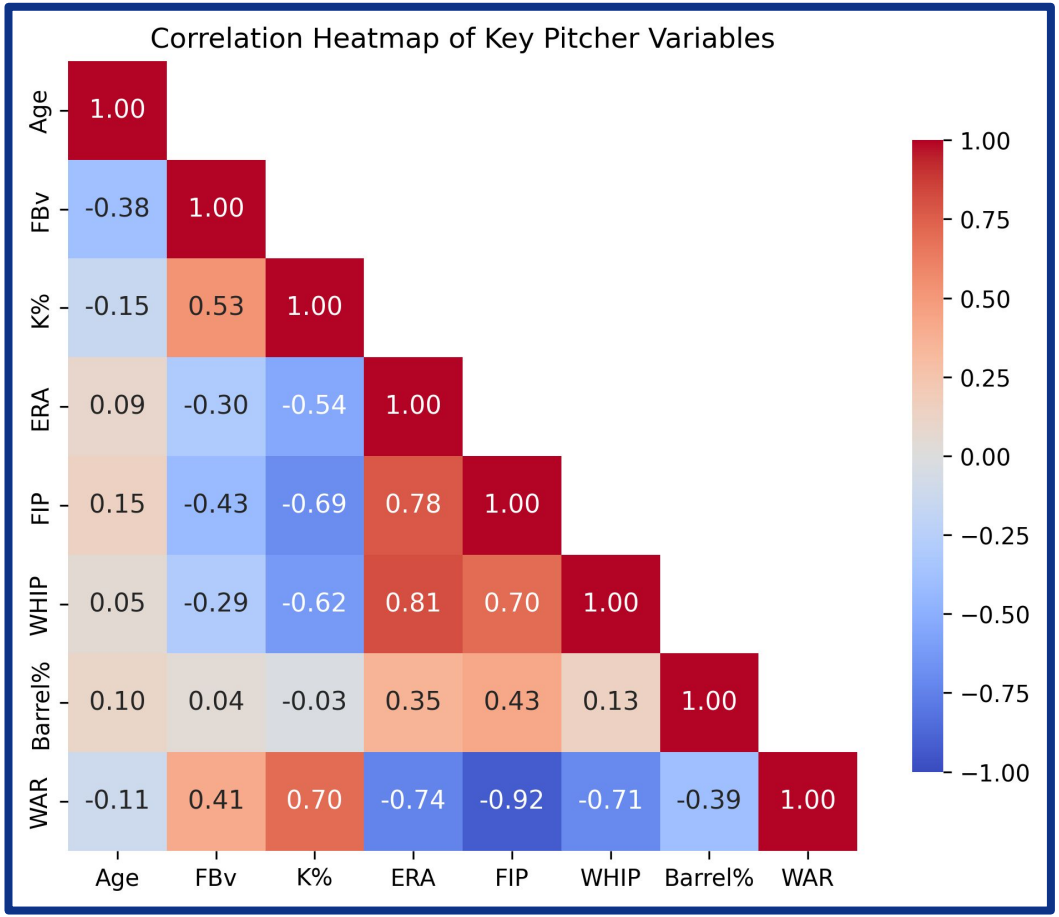
WAR by Age

- Typically the curve follows an upside down U shaped pattern which is not the case here
  - Our primary suspicion is survivorship bias
- Skubal will be 30, so this is important to keep in mind when deciding the number of years



Correlation between Variables

- Allows us to see which variables have signal and have some overlap
  - WAR has a strong negative correlation with ERA, FIP, and WHIP
  - ERA, FIP, and WHIP all have a strong positive correlation with each other
- Something to keep in mind while modeling, especially when thinking about multicollinearity



Base Modeling

Train/Test Split

Problem: Dataset contains multiple pitcher-years
Ex. Jacob DeGrom 2021, Jacob DeGrom 2022
Model will be able to memorize pitcher archetypes unless train/test are stratified



Solution: ensure that no single pitcher exists in both the train and test datasets

Standardizing

Problem: We are working with data that changes with time
Ex. 93 fastball velocity was fast in 2016, but slow in 2025
WAR is relative to the league average for that season, and using base continuous values will not reflect this



Solution: standardize all predictors for each season

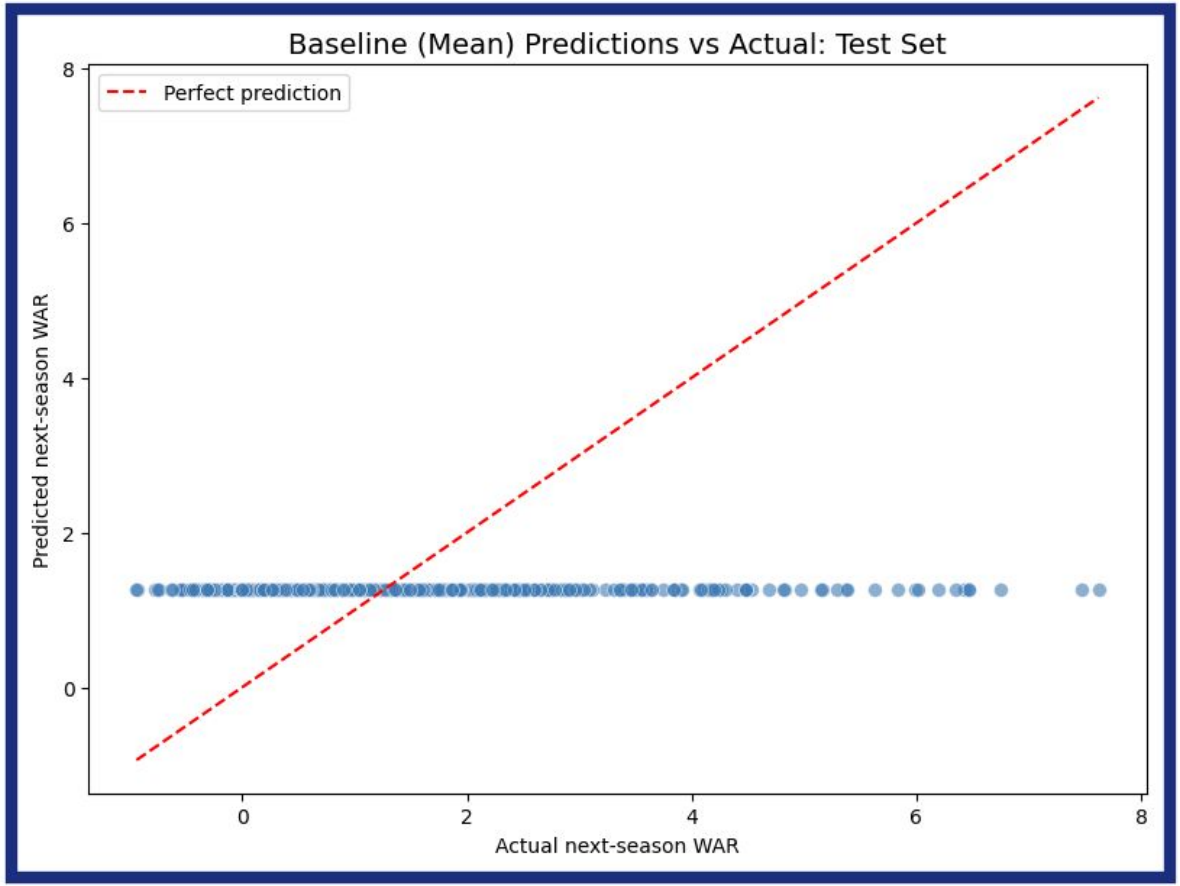
Modeling Plan So Far

- Training set consists of 1791 pitcher seasons, test set consists of 462 pitcher seasons, with no pitchers crossing over
- For our first batch of models, we decided to predict next-season WAR using current season stats. This may change in future iterations.
- Three models created
    - Decision tree, no tuning
    - Gradient Boosting Regressor, small tuning grid (can be improved)
    - Elastic Net, alpha and l1 ratio both tuned
- NA data was imputed with the median apart from fastball velocity which was imputed with 0 for the tree models



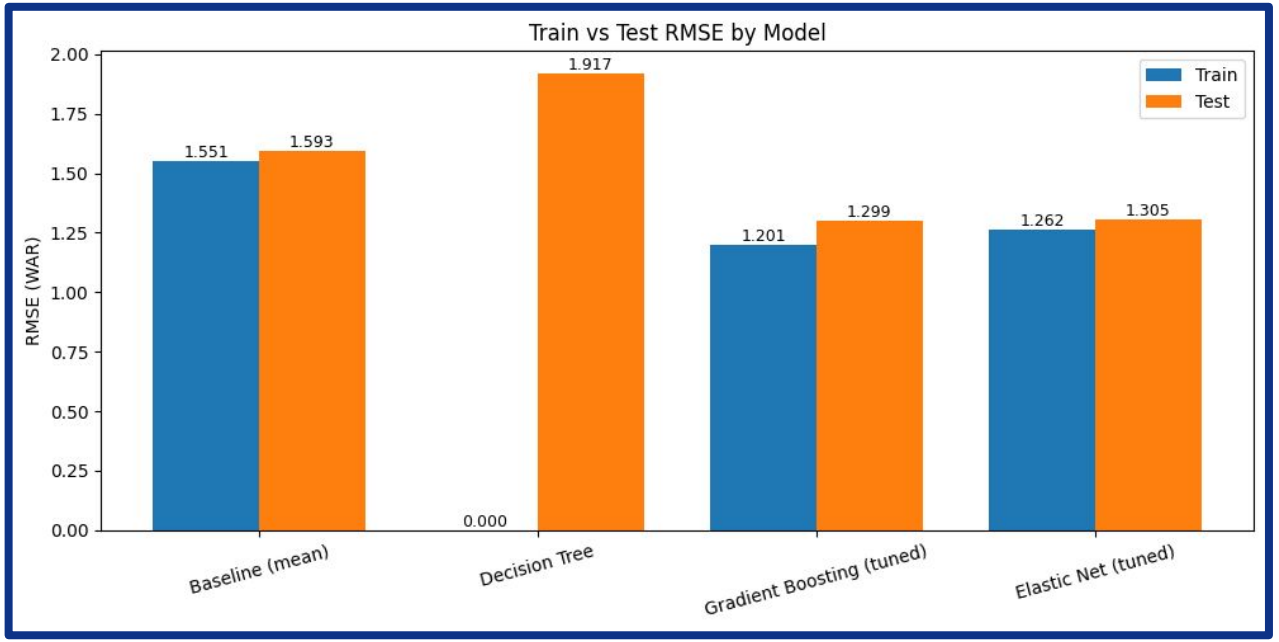
Baseline Model

- We need a baseline model to compare our RMSE to so we know how much the model actually improves predictions in comparison
- The baseline model predicts the mean value of the response from the test set for every value
    - RMSE: 1.593



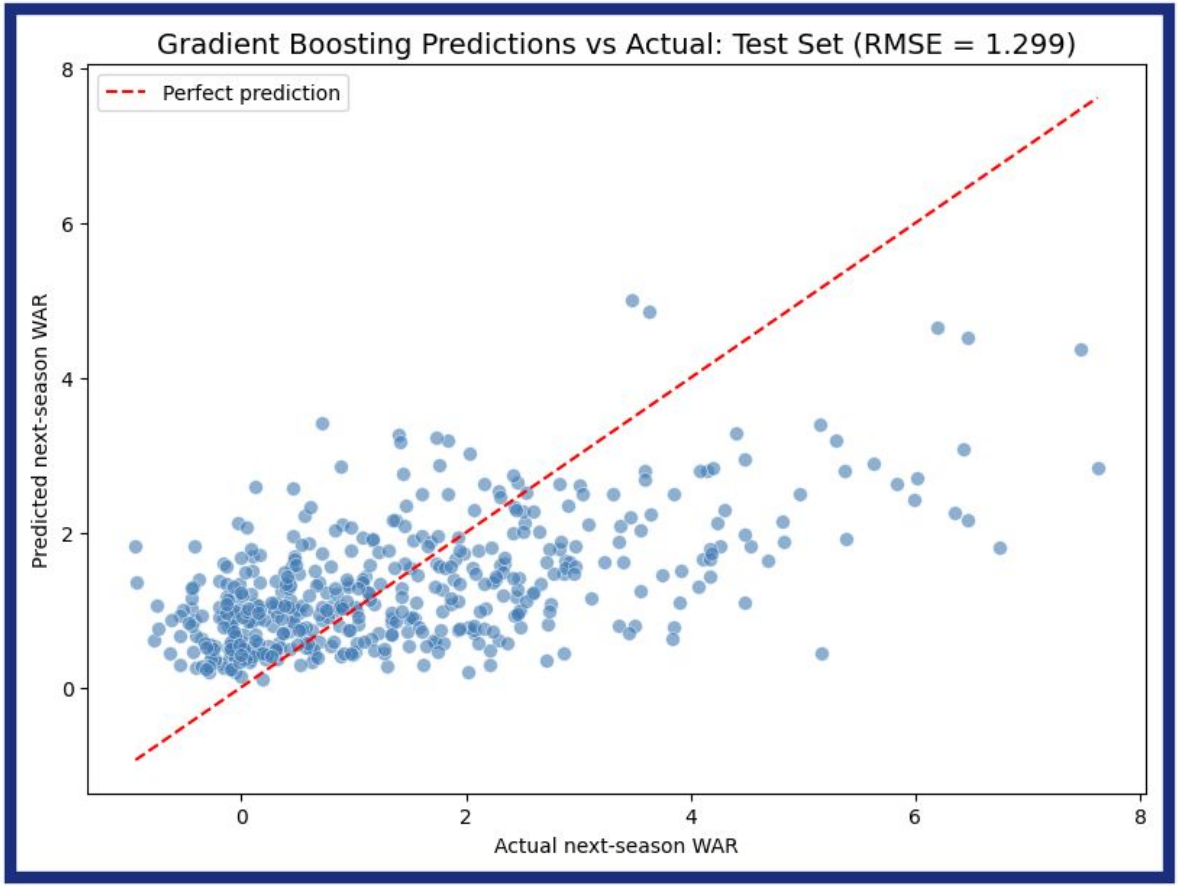
Best Model

- Decision tree badly overfits and is worse than the baseline
- Gradient boosting and Elastic Net have very similar results
- Elastic Net is the best current model



Gradient Boosting Regressor Results

- Outperforms Baseline, but not by as much as we would like
- Struggles with extreme values
- We likely need to update the next iterations of models with variables including information about playing time



Gradient Boosting Regressor Tuning

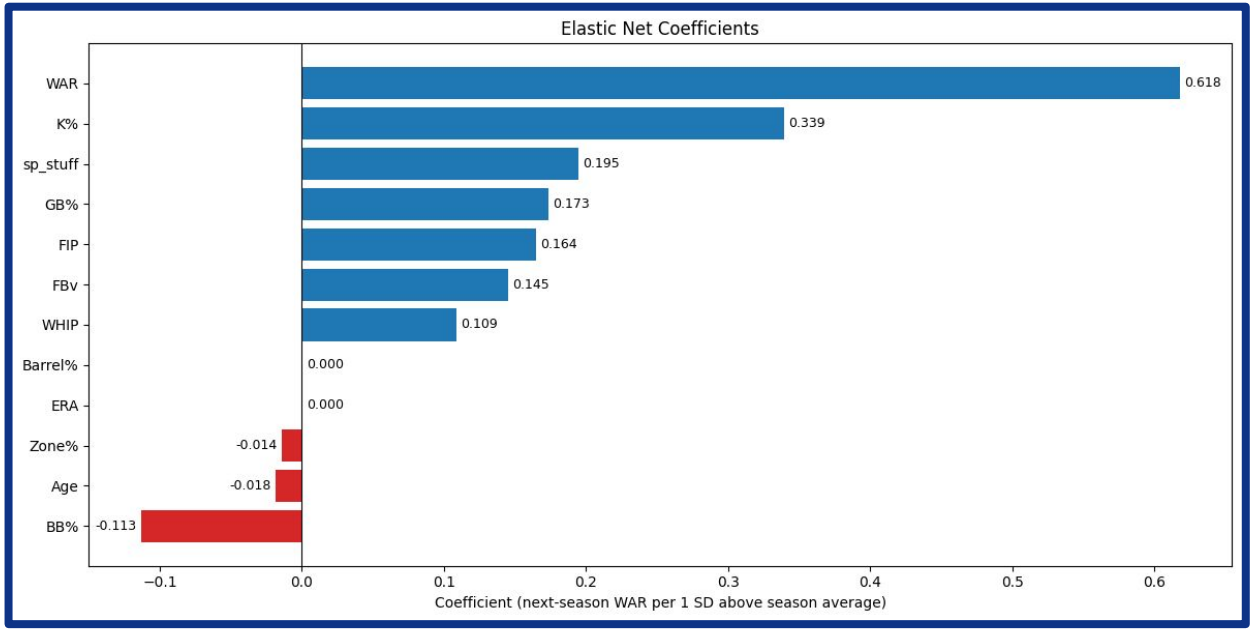
- Small tuning grid for initial test and computational efficiency
- Small learning rate wins, likely could be smaller
- More tuning can be done to refine this model

Gradient Boosting Tuning Results (best combination highlighted)

Rank	n_estimators	learning_rate	max_depth	CV RMSE Mean	CV RMSE Std
1	**100**	**0.05**	**2**	**1.296**	**0.066**
2	100	0.05	3	1.300	0.068
3	100	0.1	2	1.309	0.069
4	300	0.05	2	1.314	0.072
5	100	0.1	3	1.317	0.071
6	300	0.05	3	1.318	0.072
7	300	0.1	2	1.330	0.078
8	300	0.1	3	1.343	0.080

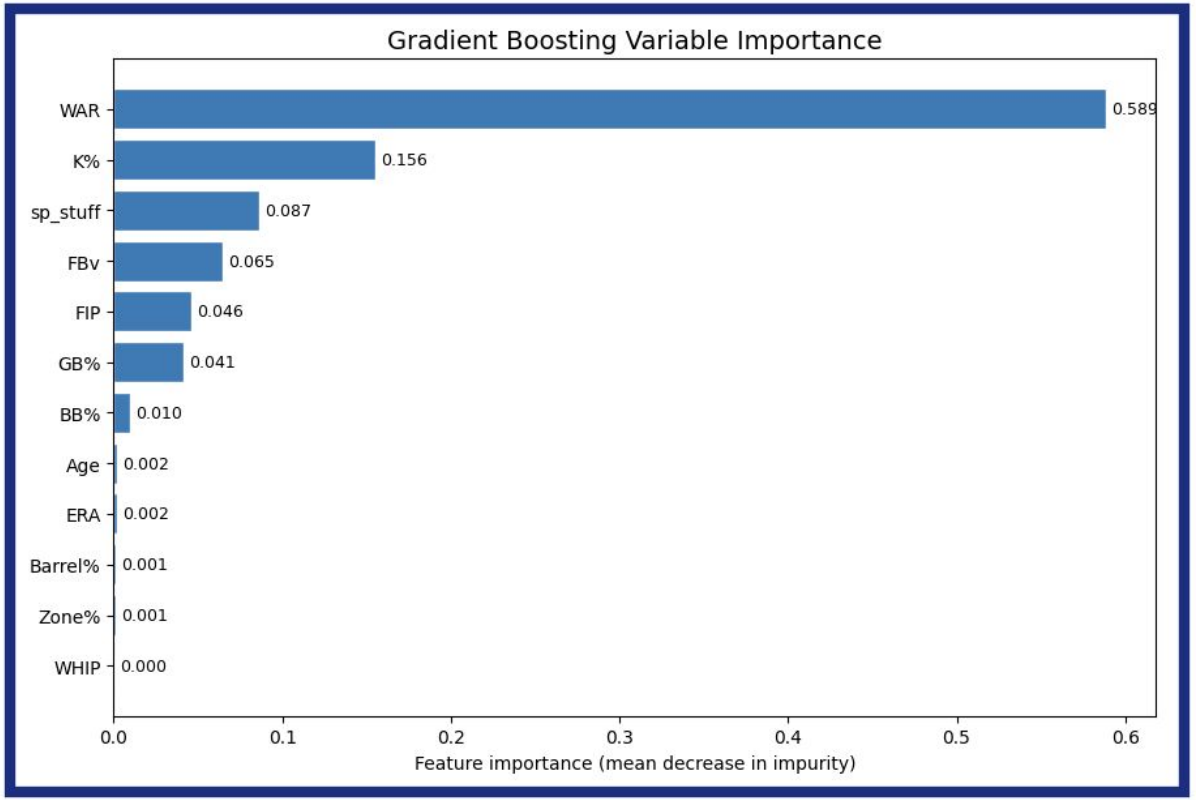
Elastic Net Coefficients

- Previous season WAR, K%, BB%, and Stuff+ have the highest impact in the Elastic Net, which makes intuitive sense



Gradient Boosting Analysis

- Gradient Boost saw similar results, but did not value BB% as much
- Model barely overperformed elastic net, but was not tuned with extreme detail. A more tuned model may further outperform Elastic Net the future iterations



Future Work/Problems

Future Work:

- We failed to consider how playing time would factor into next season projections
  - Did not initially included because it is not a performance stat, but it intuitively matters
- Tune Gradient Boosting model more: with little tuning it barely beat Elastic Net
- Create a multi-year framework: more models that predict more years into future

Challenges/Problems:

- Models learned in class are not the most intuitive for this issue:
  - Projecting multiple years into the future is best completed by complex systems, Bayesian Hierarchical models and Deep Learning models
  - For simplicity, we eventually will have to work around this by creating multiple models

Thank You!
Questions?